

AXEL LEVY

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EDUCATION

Stanford University Ph.D. Electrical Engineering - Efficient Machine Learning Algorithms for Protein Structure Determination with Cryo-EM - Areas: AI for science, generative AI, protein structure determination, 3D reconstruction	2020 - 2025
Ecole Polytechnique Grandes Ecoles Program / Ingénieur Polytechnicien Areas: statistical mechanics, quantum physics	2017 - 2020
Lycée Sainte-Genevieve MP - PSI* Areas: maths, physics, engineering sciences	2015 - 2017

RESEARCH AND WORK EXPERIENCE

Generate Biomedicines - Machine Learning Research Scientist Somerville, Massachusetts - AI for drug discovery - Frontier generative models for biomolecules - Lead on the development of an active learning loop, connecting experimental data to model training - Joined pre-IPO	2025 - Present
Generate Biomedicines - Machine Learning Intern Somerville, Massachusetts - De novo protein generation with diffusion models, structure determination - Advisor: Dr. John Ingraham, Dr. Eugene Palovcak, Dr. Maxwell Zimmerman	Jun 2024 - Sep 2024
Google Brain - Research Scientist Intern Mountain View, California - Inverse rendering, neural rendering - Advisor: Dr. Dmitry Lagun (SynthX team)	Jun 2022 - Sep 2022
SLAC National Lab + Stanford Computational Imaging Lab Stanford, California Advisors: Prof. Mike Dunne (LCLS), Prof. Gordon Wetzstein (SCI)	2021 - 2025
University of Sheffield - Visiting Researcher Sheffield, UK (remote) - Reservoir computing, neuromorphic computing, quantum materials - Advisor: Prof. Tom Hayward, Dr. Alex Welbourne	Mar 2020 - Jul 2020
ONERA - The French Aerospace Lab Palaiseau, France - Random matrices theory, diffusion theory - Advisors: Dr. Jean-Philippe Ovarlez, Dr. Eugenie Terreaux	Sep 2019 - Mar 2020
Polytechnique Scientific Group Project Palaiseau, France - Built a quantum remote laboratory in a group of 5 students https://fentes-young-photon-exp.binets.fr/index.php - Advisors: Prof. Emmanuel Joffre, Prof. Daniel Suchet	Sep 2018 - Jun 2019
Stereolabs - Optoelectronic Engineering Intern Paris, France	Jun 2019 - Aug 2019

- 3D stereo-imaging, optics, electronics
- Start-up of 15 people
- *Advisor: Dr. Edwin Azzar*

Air Force Officer

Noumea, New Caledonia

2018

PUBLICATIONS

CryoDRGN-AI: Neural *Ab Initio* Reconstruction of Challenging Cryo-EM and Cryo-ET Datasets 2025

Levy A., Raghu R., Feathers R., Grzadkowski M., Poitevin F., Johnston J., Vallese F., Clarke O., Wetzstein G., Zhong E.

Nature Methods - [link](#)

Multiscale Guidance of AlphaFold3 with Heterogeneous Cryo-EM Data (CryoBoltz) 2025

Raghu R., Levy A., Wetzstein G., Zhong E.

NeurIPS - [link](#)

Scalable 3D Reconstruction for X-Ray Single Particle Imaging Based on Online Machine Learning 2025

Shenoy J., Levy A., Ayyer K., Poitevin F., Wetzstein G.

Nature Communications

Mixture of Neural Fields for Heterogeneous Reconstruction in Cryo-EM 2024

Levy A., Raghu R., Shustin D., Peng A., Li H., Clarke O., Wetzstein G., Zhong E.

NeurIPS - [link](#)

Towards Interpretable Cryo-EM: Disentangling Latent Spaces of Molecular Conformations 2024

Klindt D., Hyvärinen A., Levy A., Miolane N., Poitevin F.

Frontiers in Molecular Biosciences - [link](#)

MELON: NeRF with Unposed Images in SO(3) 2023

Levy A., Matthews M.*, Sela M., Wetzstein G., Lagun D.*

3DV Spotlight - [link](#)

GeNVS: Generative Novel View Synthesis with 3D-Aware Diffusion Models 2023

Chan E., Nagano K.*, Chan M.*, Bergman A.*, Park J.J.*, Levy A., Aittala M., De Mello S., Karras T., Wetzstein G.*

ICCV Oral - [link](#)

Amortized Inference for Heterogeneous Reconstruction in Cryo-EM 2022

Levy A., Wetzstein G., Martel J., Poitevin F., Zhong E.

NeurIPS - [link](#)

Heterogeneous reconstruction of deformable atomic models in Cryo-EM 2022

Nashed Y., Peck A., Martel J., Levy A., Koo B., Wetzstein G., Miolane N., Ratner D., Poitevin F.

NeurIPS Workshops - [link](#)

CryoAI: Amortized Inference of Poses for *Ab Initio* Reconstruction of 3D Molecular Volumes from Real Cryo-EM Images 2022

Levy A., Poitevin F., Martel J., Nashed Y., Peck A., Miolane N., Dunne M., Wetzstein G.

ECCV - [link](#)

Deep Generative Modeling for Volume Reconstruction in Cryo-Electron Microscopy 2022

Donnat C., Levy A., Poitevin F., Zhong E., Miolane N.

Journal of Structural Biology - [link](#)

Voltage-controlled superparamagnetic ensembles for low-power reservoir computing 2021

Welbourne A., Levy A., Ellis M., Chen H., Thompson M., Vasilaki E., Allwood D., Hayward T.

Applied Physics Letters - [link](#)

TALKS

PhD Defense (Stanford University, Stanford) <i>recorded</i>	2025
Guest Lecture (<i>Laura Waller</i> , UC Berkeley, Berkeley)	2024
SLAC Users' Meeting (SLAC, Menlo Park)	2024
Guest Lecture (<i>Gordon Wetzstein</i> , Stanford University, Stanford)	2024
SCIEN Seminar (Stanford University, Stanford) <i>recorded</i>	2023
CECAM Workshop (EPFL, Lausanne)	2023
Invited Talk (<i>Slavica Jonic</i> , IMPMC, Paris)	2023
Invited Talk (<i>Michael Unser</i> , EPFL, Lausanne)	2023
Invited Talk (<i>George Drettakis</i> , INRIA, Sophia-Antipolis)	2023
Invited Talk (Google, Mountain View)	2023
Invited Talk (<i>Vincent Sitzmann</i> , MIT, Cambridge)	2023
SLAC Public Lecture (SLAC, Menlo Park) <i>recorded</i>	2023
Invited Talk (<i>Ellen Zhong</i> , Princeton University, Princeton)	2022
Invited Talk (<i>Pilar Cossio</i> , Flatiron Institute, NYC)	2022
Invited Talk (<i>Sylvain Gigan</i> , ENS, Paris)	2022
SLAC AI Seminar (SLAC, Menlo Park)	2022
PIMS Workshop (UBC, Vancouver)	2022
4th International Symposium on Cryo-3D Image Analysis (Tahoe)	2022

AWARDS AND FELLOWSHIPS

Google Conference Scholarship	2024
Pacific Institute for the Mathematical Sciences Travel Award	2022
SGF: Stanford Graduate Fellowship in Engineering	2020
French Academy of Sciences Prize (awarded for scientific work at Ecole Polytechnique)	2021
Best Scientific Group Project Prize (awarded to 3 groups out of 120)	2020

ACADEMIC SERVICE

Reviewer, European Conference on Computer Vision (ECCV)	2024
Reviewer, Neural Information Processing System (NeurIPS)	2023, 2024, 2025
Reviewer, NeurIPS Workshop on Machine Learning for Structural Biology (MLSB)	2023, 2024
Reviewer, International Conference on Computer Vision (ICCV)	2023
Reviewer, Conference on Computer Vision and Pattern Recognition (CVPR)	2022, 2024
Reviewer, PLOS Computational Biology	2022

TEACHING EXPERIENCE

Course Assistant - Computational Imaging	2023 - 2024
<i>Instructor: Pr. Gordon Wetzstein (Stanford University)</i>	
Course Assistant - Semiconductor Memory Devices and Circuit Design	2020
<i>Instructor: Pr. Philip Wong (Stanford University)</i>	
Teaching Assistant in Preparatory School - Physics	2018 - 2019
<i>Instructor: Pr. Thibaud Naulet (Ecole Sainte-Genevieve)</i>	
• Prepared students for a nationwide ranked exam (X/ENS/Centrale/Mines)	

SOFTWARE

ADP-3D: https://github.com/axel-levy/axlevy-adp-3d	2024
DRGN-AI: https://github.com/ml-struct-bio/drgnai	2024
CryoFIRE: https://github.com/ml-struct-bio/cryofire	2022
CryoAI: https://github.com/compSPI/cryoAI	2022

SKILLS

Languages: English (fluent), French (native), Spanish (intermediate)

Programming: Python, Pytorch, Jax, Tensorflow, C++, Java, Git